Project #5

System Integration and Testing

Stein Lysberg

May 1, 2026

Contents

1 Introduction

This report is the fifth in a series of five. It will look at use cases for the scanner, the system architecture, system integration and testing of the finished plastic scanner. As the plastic scanner finishes a scan, the results are sent to a computer via serial communication. In the computer, these reading are presented on the monitor via a graphical user interface. This user interface has the option to store data, which allows for the creation of a database of values for known materials. Using this database, the interface could compare any value to already known values, and have a guess at which material it is scanning.

1.1 Use cases

Because the plastic scanner requires a computer to present its output, it is not very portable. This report will show that it is also sensitive to changing environments. For this reason, a realistic use case for could be at a stationary sorting station for plastic waste, shielded from sunlight. For example the the back of a van during a beach cleaning, or at a makeshift sorting facility. It should also be noted that the scanner is not yet reliable to distingush between all types of plastic, but it could still be useful for coarse sorting.

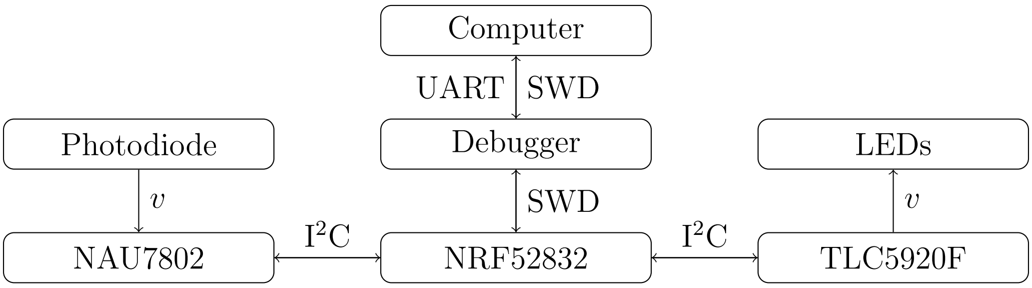


Figure 1: System architecture

1.2 System architecture

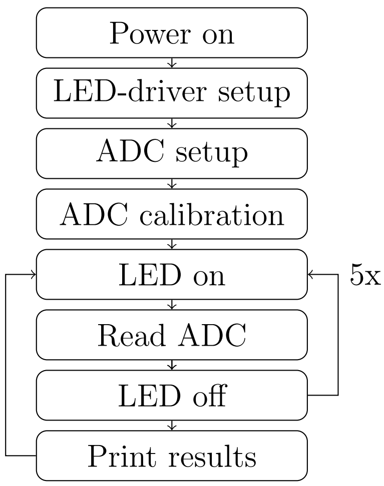


Figure 2: Workflow chart

Reports 3 and 4 discussed the use of the TLC5920F LED-driver and the NAU7802 ADC individually. This report will look at how they work together as part of the system. Figure 1 shows the system architecture. The computer programs the NRF52832 microcontroller using Serial Wire Debug protocol, via the debugger on the development kit. The debugger is also a NRF52832, but in this application it only handles interfacing with the computer. The microcontroller controls both the ADC and the LED-driver using I^2C. Led LED-driver supplies voltage to the LEDs. The ADC reads a voltage from the photodiode.

Figure 2 shows the flow of the code in this project. After power-up, the microcontroller checks that it can communicate with the LED-driver. it then configures the LED-driver by writing the appropriate values to its registers. The process is repeated for the ADC, which is also prompted to do its own internal calibration routine. Following this, the code loops through the LEDs by turning them on, reading from the ADC, and turning them off again. After all six LEDs have been looped over, the results from the six scans are printed to the console.

2 Testing

Tests have been conducted by performing scans of various items and environments, and then storing and analyzing the results. These tests will be explained in the following section. The lines on the graphs are the averages of a series consisting of between 20 and 100 readings. The shaded areas around the lines are the averages plus/minus one standard deviation. The x-axis represent the wavelength of the LED that is currently on, and the y-axis is normalized between the minimum and maximum values across all readings we have done.

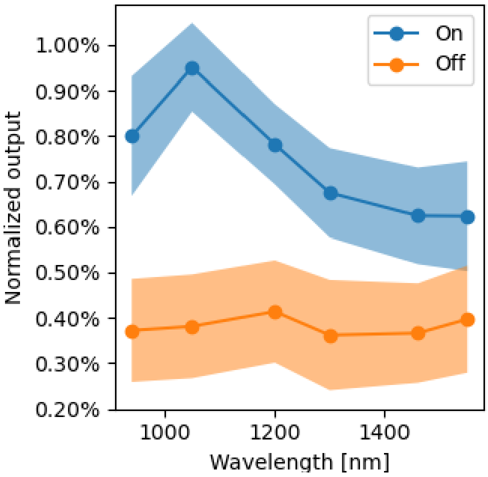


Figure 3: Effect of LEDs.

2.1 Integration testing

Figure 3 shows the effect of the LEDs on the ADC. For this dataset the plastic scanner sat on a table making readings of the air, with the LEDs either on or off, and approximately 100 readings were taken in both cases.

2.2 Scanning in- and outdoors

Figure 4 shows the effects of gain on the readings, as well as the effect of environmental light. The tests were conducted indoors and outdoors, using the maximum gain for the ADC, which is 128, and the minimum which is 0. The indoors tests were conducted in a room with the lights on and the curtains closed, and the outdoors tests were conducted in the window sill of the same room, where the photodiode sensor was exposed to the sun. In this scale, the bottom two sets of readings (indoors with and without gain) are almost indistinguishable. 70-100 readings were taken for each line. This graph also has a shaded area representing plus/minus one standard deviation, like fig. 3, but the scale is so large that they are indistinguishable from the lines.

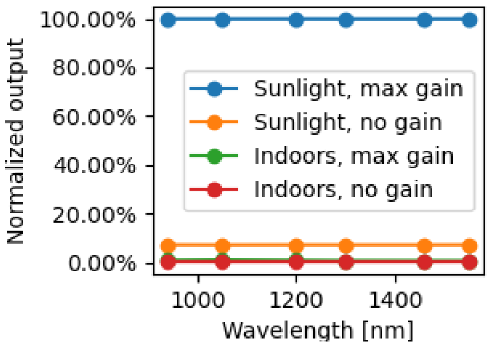


Figure 4: Effect of sunlight and gain.

2.3 Scanning production pellets

Figure 5 shows the results of some scans of plastic production pellets in a small box with mirrored walls. In this experiment the box sat on the table, was filled with pellets, and

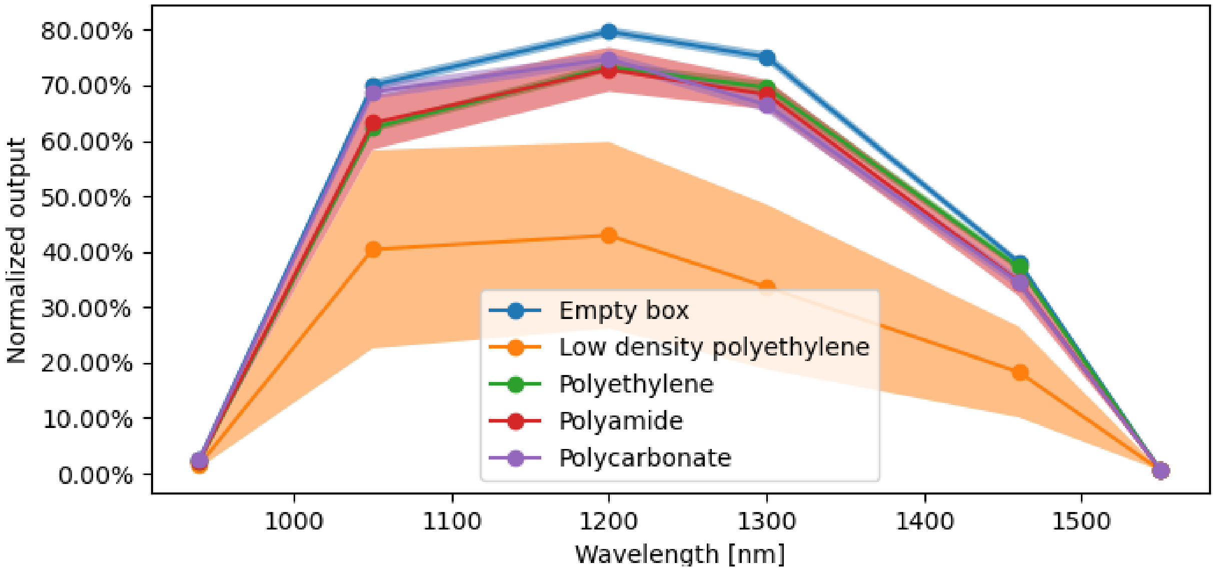
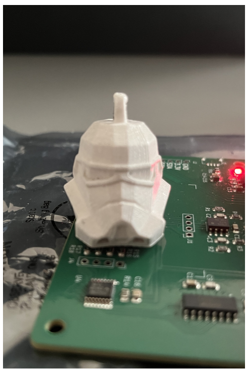


Figure 5: Scans of various production pellets.

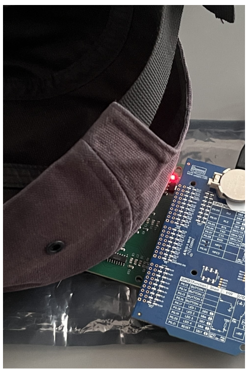
the plastic scanner was held over it to scan into the box. The top line is the box with mirrored walls, without contents, so as to reflect as much light as possible, included for reference. 20-30 readings were taken for each material.

2.4 Scanning items

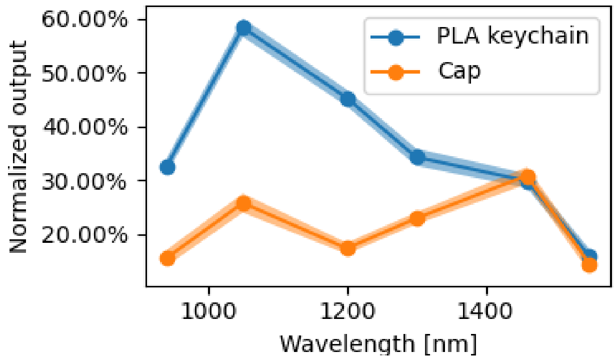
In addition to scanning environments and pellets, a few items were scanned to see how the scanner distinguishes between them. Namely, a polyester cap and a 3D-printed PLA keychain. As fig. 6 shows, the dinstinction is clear, showing that the plastic scanner can make clear distinctions between different objects in the same environment. 30 readings were taken for the cap, and 50 for the keychain.



(a) Keychain.



(b) Cap.



(c) Spectra.

Figure 6: A cap, a keychain, and their spectra.

3 Discussion

This section will discuss the results from the tests, as well as suggest some ways to further improve upon the current state of the plastic scanner.

3.1 Test results

3.1.1 Integration

As shown in fig. 3, the LEDs being on affect the values from the ADC, which indicates that these components work together as they should. Since these reading were taken without an object in front of the scanner, these results also show that the photodiode sensor picks up some light directly from the LEDs.

3.1.2 Environments

Figure 4 showed the effect of gain on the values produced by the ADC, as well as the environment light. The scan of sunlight shows all readings close to 100%, which means they are the highest readings recorded in any of the tests, because the scale is normalized between lowest and highest recorded value across all tests. It is worth noting that a scan in the sunlight with no gain yields higher readings than an indoor scan with maximum gain.

3.1.3 Materials and items

Figure 5, the results of scans of production pellets made of various plastics, shows that different items yield different readings. The empty box is easily distinguishable from low density polyethylene, but regular polyethylene, polyamide and polycarbonate look about the same. as shown in fig. 7. The cap and keychain in fig. 6 shows that the scanner has the potential to clearly distinguish different items, but it is unclear whether this is because they are made from different plastics, or because they differ in some other way, like coloring or density.

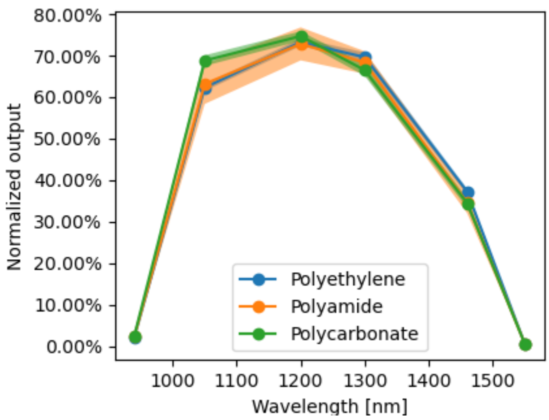


Figure 7: Excerpt from fig. 5.

3.1.4 Issues and sources of error

Not being able to distinguish between polyethylene, polyamide and polycarbonate, as shown in fig. 5, is a problem. The cause for this could be that the box was not sufficiently full, so the readings were more affected by the mirrors than the pellets. This could also why these curves are so close to that of the empty box.

Another source of error in this experiment is that the environment may not have been the same in all tests. The plastic scanner was held manually as the scan was conducted, so alignment and spacing with the box may not have been the same for all readings. This is especially noteworthy in the case of low density polyethylene, where the standard deviation is greatest.

3.2 Potential improvements

Although no more work will be done on the plastic scanner, a few potential improvements to it will be suggested in the following section. They are separated into potential software and hardware improvements.

3.2.1 Software improvements

The suggested improvement or using interrupts rather than polling for the ADC was discussed in report 4. This suggestion will build on that. The code for the plastic scanner loops continuously, starting the next scan cycle immediately after printing the previous. Instead of doing this, a scan cycle (or a number of scan cycles) could be triggered by a button connected to the GPIO of the nRF52. If this was implemented, the nRF52, TLC5920F and NAU7802 could all be set to low-power mode between the scans. This improvement would be most relevant if this was to be used as a handheld device. This could be similar to the command line interface implementation in the Arduino code [1].

As fig. 4 shows, the readings vary a lot with the gain and environment lighting. A function could be added, also triggered by a switch connected to the GPIO, which sets an appropriate gain for the ADC. This would let readings be more similar across different environments. It could be implemented as shown in fig. 8. Note that the ADC has to do its own calibration routine after setting a new gain [2].

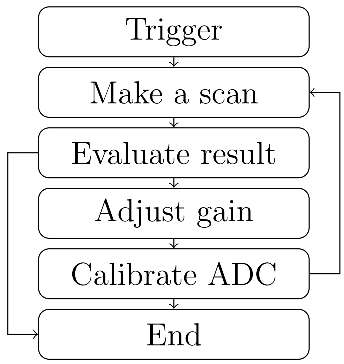


Figure 8: Gain calibration.

3.2.2 Hardware improvements

To increase the reliability of the scanner, more LEDs could be added for increased resolution. The LEDs currently on the PCB cover the full spectral range of the photodiode sensor, but there could be *more* LEDs within this range, creating a finer resolution. The two idle LED slots on the PCB could be used for this. To further increase reliability, more and more leds and photodiodes with different wavelengths and spectral ranges can be added, but it should be kept to a minimum to make sense in the contect of this project.

As fig. 3 shows, the photodiode picks up some light directly from the LEDs. If this could be reduced, more of the collected light would be *reflected* light, which is what is useful. This could be achieved by making a case for the plastic scanner, with a physical barrier between the LEDs and the photodiode. A case would also make the whole contraption less vulnerable, and nicer to handle.

4 Conclusions

This report has shown that although the plastic scanner in its current state produces appropriate results in the conditions it has been tested in, it is not yet fully reliable as a tool to distinguish between all different types of plastic. Although this project is finished, there are many improvements which can still be made. With more testing in different environments, and a database of known values for known materials, it may become easier to determine what type of plastic is being scanned. This may also require some knowledge of chemistry and data analysis which is beyond the scope of this report.

References

[1] J. de Vos, “DB2.x Firmware,” 2022, last accessed 13 May 2025. [Online]. Available: https://github.com/Plastic-Scanner/DB2.x-Firmware

[2] Nuvoton, “NAU7802,” 2023, last accessed 13 May 2025. [Online]. Available: https://www.nuvoton.com/export/resource-files/en-us--DS_NAU7802_DataSheet_EN_Rev2.6.pdf